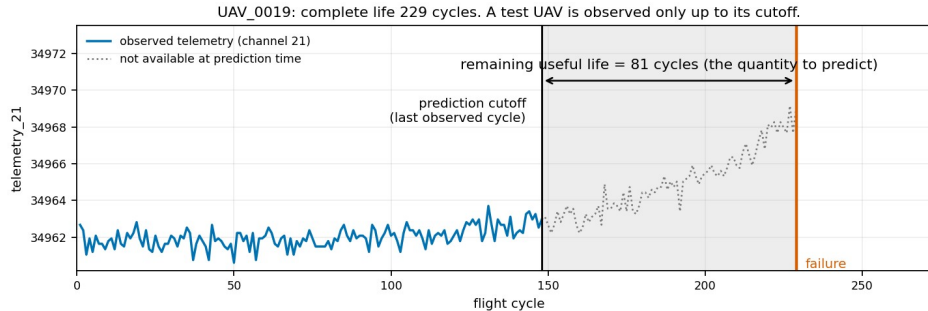


University of Stuttgart
Germany

Estimating remaining useful life from a partial UAV telemetry history



University of Stuttgart · 9 September 2026

Figure: generated for this talk from data/train.csv · claims C01-C04

How we reached a public R^2 of 0.87652

Joest
Homann

Five submissions, five decisions

Each point is a submitted model; the label names the decision that run introduced

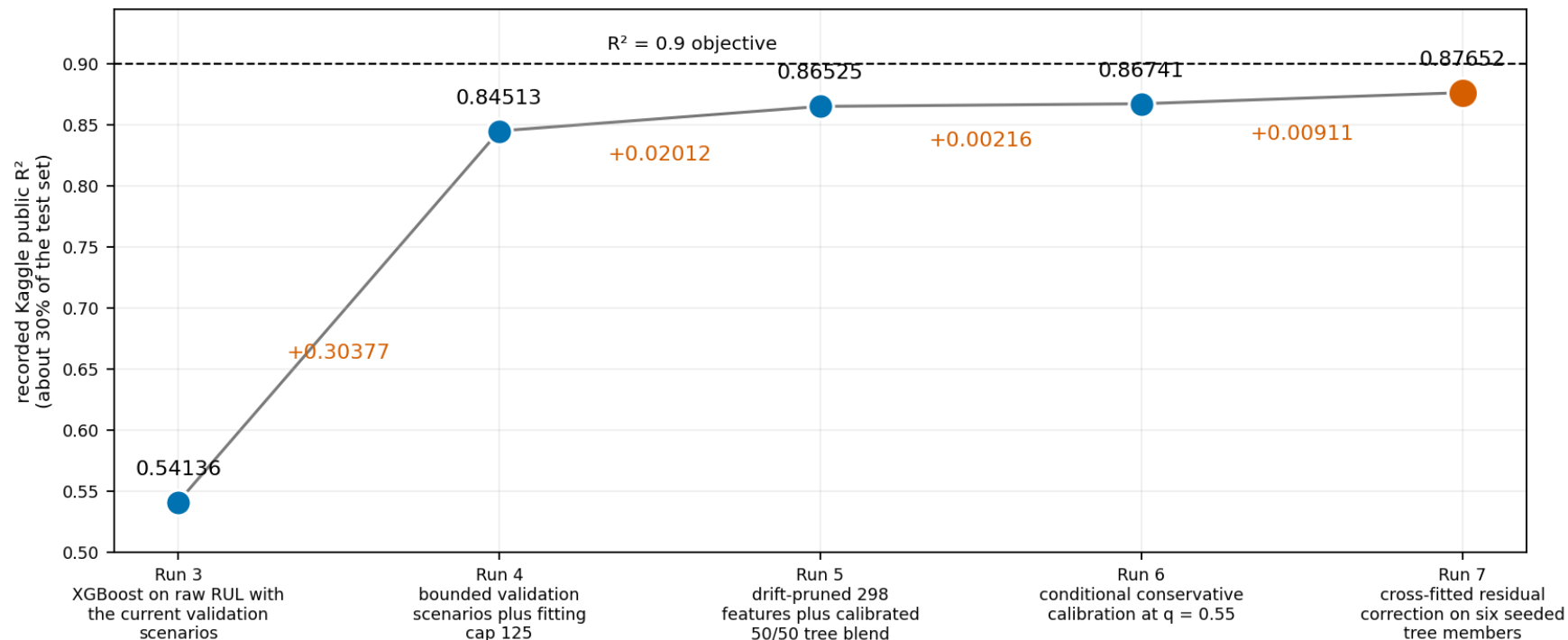


Figure: generated from kaggle_scores.csv and pipeline_experiments.md · claims C05-C12

Six of twenty-eight channels carry no information

Red: unique-value count and numeric range at the effectively-constant threshold

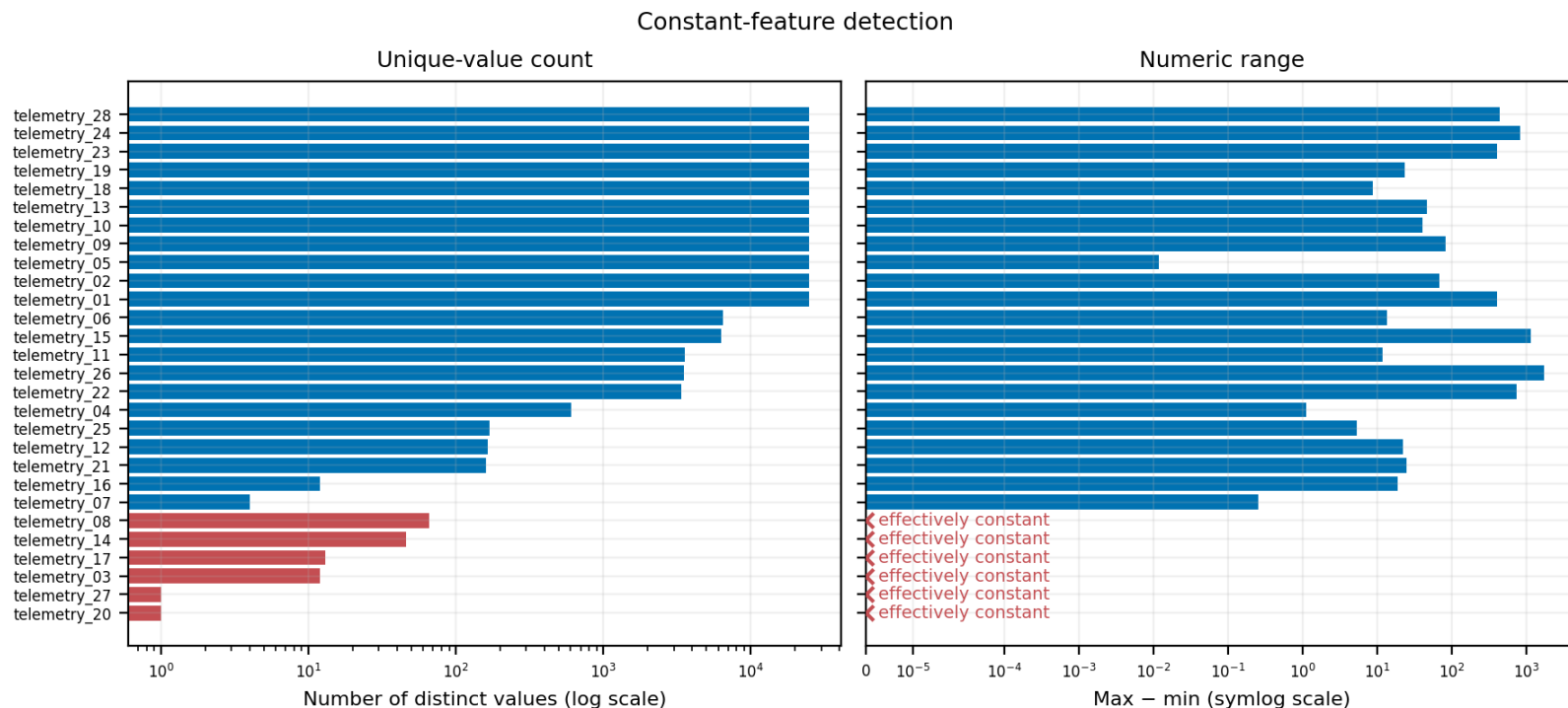


Figure: repository script re-run at slide size · 0_data_analysis/broad_data_review/plot_constant_features.py · claims C13-C18

Ten channels move with age inside every UAV

Twelve of the twenty-two channels: the ten degradation candidates, plus telemetry_18 and 26

Temporal and RUL telemetry evidence

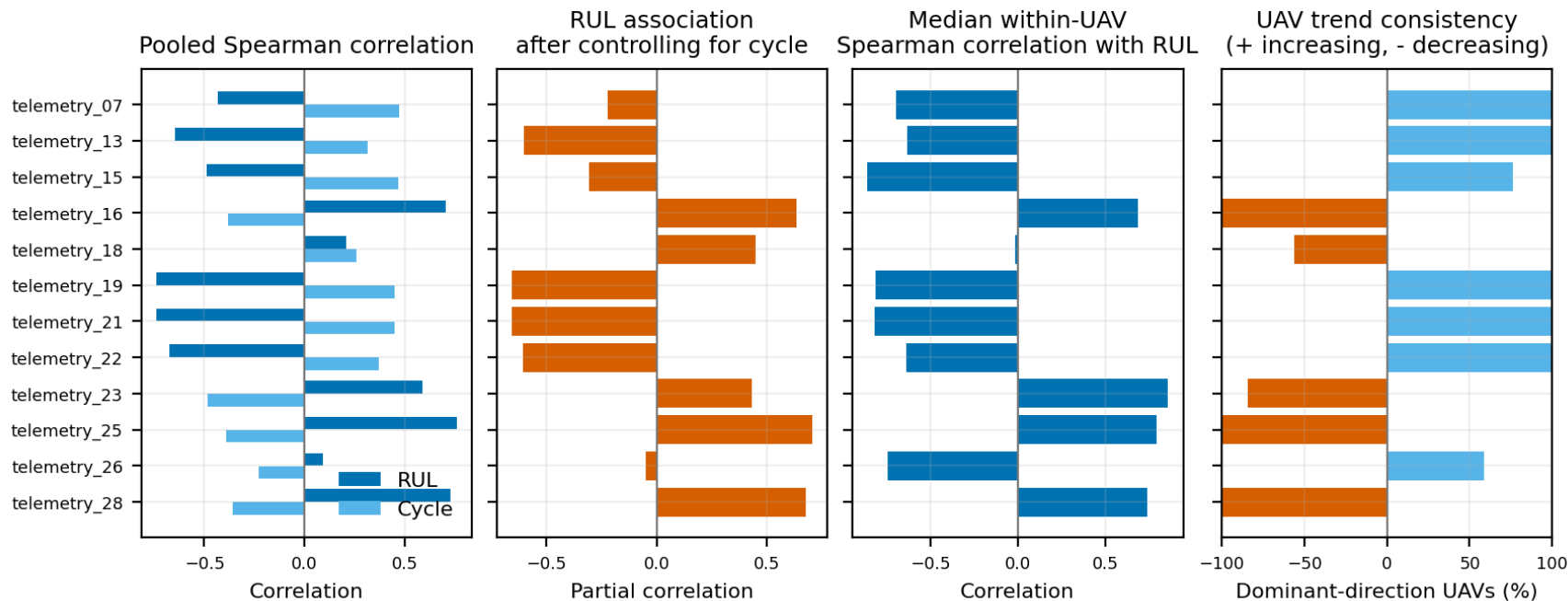
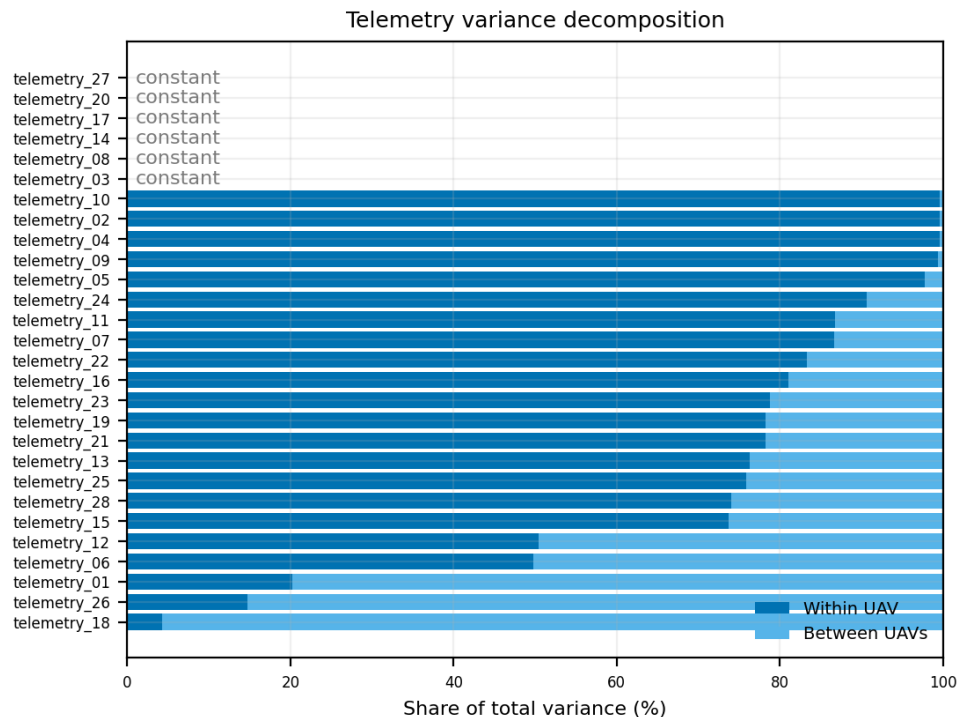


Figure: repository script re-run on a channel subset · 0_data_analysis/core_data_analysis/temporal_rul_analysis.py · claims C19-C27

Variance decomposition assigns each channel its role

Dark: variation inside one UAV over time. Light: persistent differences between UAVs.



Within-UAV dominated → sequence features

- Seventeen channels, including all ten degradation candidates (74-87% within).
- Get slope, rolling statistics, recent change and deviation from the UAV's own early-life baseline.

Between-UAV dominated → context features

- telemetry_18 (95.7% between), 26 (85.3%), 01 (79.8%), 06 (50.2%).
- Kept as level and baseline descriptions only. Their relationship must be shown to generalise to unseen UAVs.

What the plot does not settle

- Within-UAV variation may be degradation, operating state or noise; between-UAV variation may be useful context or UAV identity. Only grouped validation separates them.

Figure: repository script re-run at slide size · 0_data_analysis/broad_data_review/plot_within_between_variance.py · claims C28-C34

Train and test differ in age, not in distribution

Left two panels: raw shift. Right two: the same comparison at matched flight cycle.

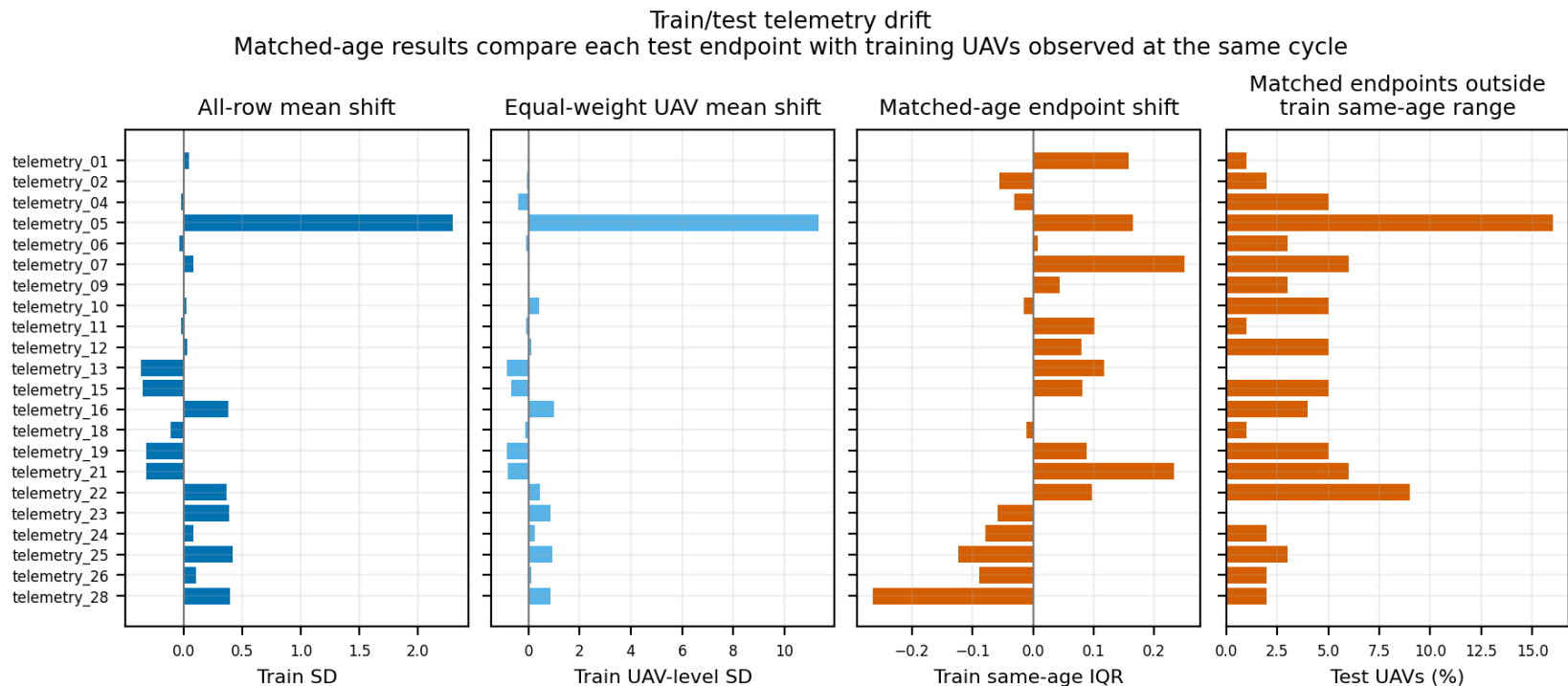
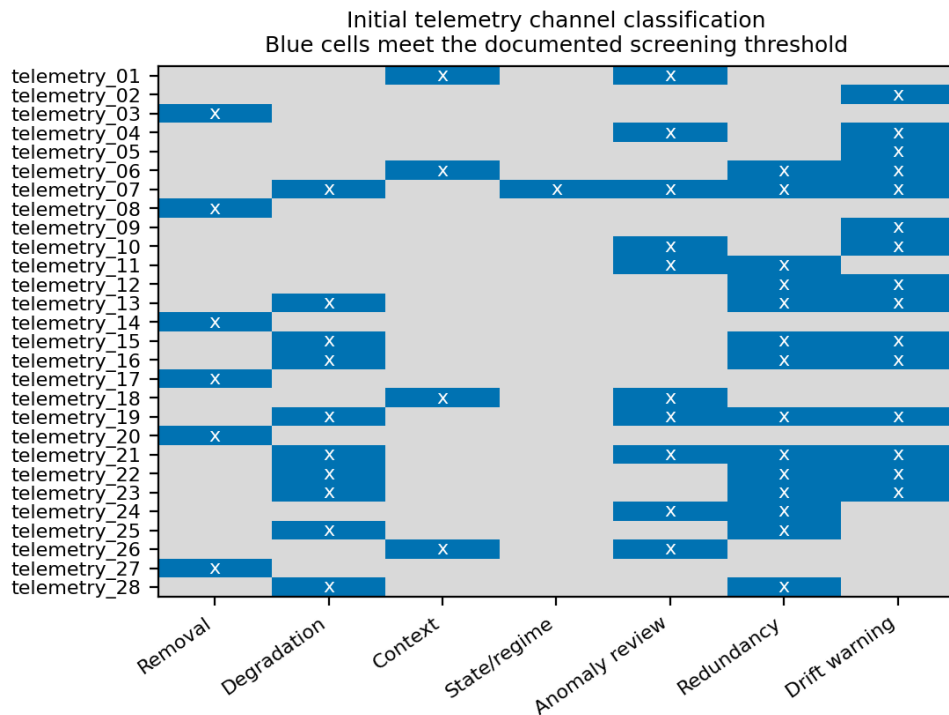


Figure: repository script re-run at slide size · 0_data_analysis/core_data_analysis/train_test_drift.py · claims C35-C41

The screening matrix that Phase 1 inherited

A blue cell means the channel meets that documented threshold. Roles are not exclusive.



What Phase 0 handed over

- 6 removal · 10 degradation · 4 context · 1 state/regime
- 10 flagged for anomaly review, 14 for redundancy, 15 for a train/test drift warning.

Warnings, not removal rules

- Four redundancy groups: 19/21, 15/23, 13/16/22/25/28 and 06/07/11/12/24. telemetry_19 and 21 correlate at $r \approx 1.00$ between UAVs.
- No channel was dropped for redundancy, anomalies or drift here. Those are questions for grouped validation.

telemetry_07 meets five criteria at once

- Degradation, state, anomaly, redundancy and drift. It is treated as an operating state with transition and dwell features, not as a smooth sensor.

Figure: repository script re-run at slide size · 0_data_analysis/core_data_analysis/channel_classification.py · claims C42-C50

Phase 1 fixed what every later number is measured against

Whole-UAV nested cross-validation, and ten automated assertions that must pass

100 training UAVs · 5 outer folds, balanced by terminal lifetime

fold 1 20 UAVs	fold 2 20 UAVs	fold 3 20 UAVs	fold 4 20 UAVs	fold 5 20 UAVs
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Outer round 1: fold 1 held out, folds 2-5 are the 80 training UAVs. Every UAV is held out exactly once.

Inside those 80 UAVs · 4 inner folds select everything

inner 1 60 train / 20 val	inner 2 60 train / 20 val	inner 3 60 train / 20 val	inner 4 60 train / 20 val
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Features, hyperparameters, early stopping and blend weights. $5 \times 4 = 20$ inner rounds.

Rules that follow from splitting by UAV

- Every prefix of one UAV stays inside one fold.
- Features use only cycles at or before the cutoff.
- Scalers are refitted inside each training partition.
- Each UAV carries a total training weight of 1.
- Uncertainty resamples whole UAVs, never rows.

Ten assertions re-checked automatically

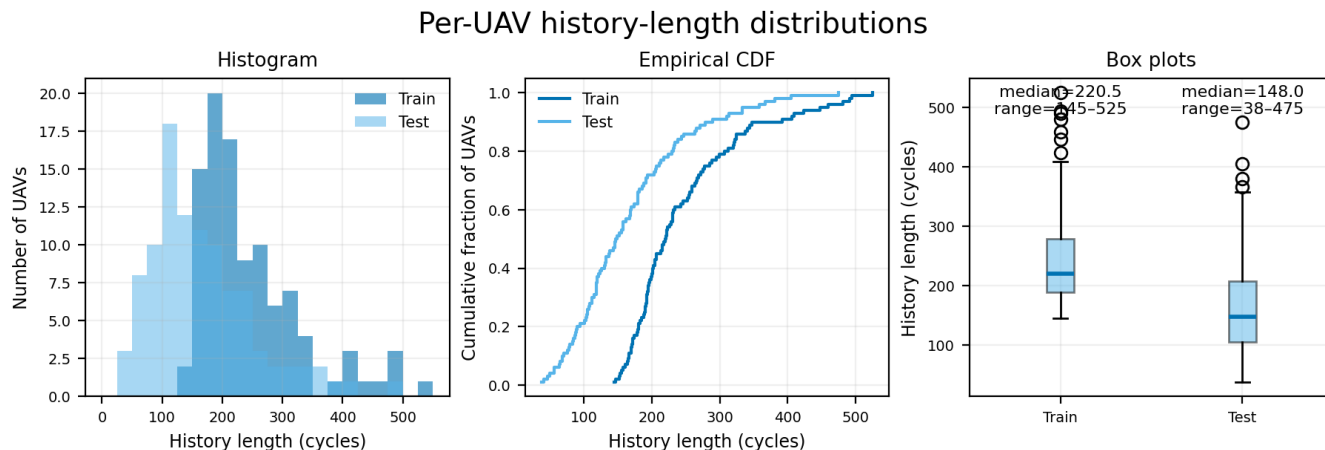
- Outer folds disjoint: 5×20 UAVs.
- Outer-validation UAVs absent from every inner fold.
- Development and locked cutoffs reproduce the exact test history-length multiset.
- Feature tables finite; no target or future term in any feature name.
- Prefix causality: replacing all post-cutoff telemetry with extreme values leaves the prefix features unchanged.
- Saved scaler parameters match a fresh recomputation.
- Baseline predictions carry each UAV's own held-out fold.

verification_report.json · status: passed · 2,000 training rows · 500 development and 2,000 locked validation rows · 100 test rows · 606 features

Sources: 1_dataset_construction/2_UAV_grouped_validation_folds and 10_automated_leakage_checks/artifacts/verification_report.json · claims C51-C59

Cutoffs are drawn from the observed test history lengths

Test UAVs are younger than training UAVs, so training samples must be truncated the same way



What the distributions say

- Train 145-525 cycles, median 220.5.
- Test 38-475, median 148.0.
- About 47% of test UAVs end before cycle 145, the shortest complete training lifetime.

How a training sample is made

- 20 distinct cutoffs per UAV, sampled without replacement from the empirical test lengths, eligible ones only.
- Each prefix uses cycles 1..cutoff and carries weight 1/20, so every UAV has total training weight 1.

What that produces

- 2,000 training rows, 5 development scenarios and 20 locked scenarios of 100 UAVs each.
- Every scenario reproduces the exact multiset of test history lengths; the check is one of the ten assertions.

Figure: repository script re-run at slide size · 0_data_analysis/broad_data_review/plot_history_length_distributions.py · claims C60-C68

606 prefix features, and four sets to compare them with

Every feature is computed from cycles at or before the cutoff, and scaled inside each fold

27 features per channel, from the prefix only

- current and first value
- baseline mean of the first 10 cycles, and the deviation from it
- history mean, SD, minimum, maximum, slope
- latest change, mean and maximum absolute change
- mean, SD, slope, net change and latest-value deviation over the last 5, 20 and 50 cycles

$22 \times 27 + 10 + 2 = 606$ candidates

- 10 state features for telemetry_07 and 16
- 2 age features
- no feature may reference terminal lifetime or any post-cutoff cycle

feature set	count	contents	question it answers
age_only	2	flight cycle and $\log(1 + \text{cycle})$	is age alone predictive?
last_values	24	age plus the latest value of each channel	does the current snapshot help?
screened	310	all temporal features of the 10 degradation channels, 7 level statistics of the 4 context channels	do the Phase 0 roles help?
all_nonconstant	606	every generated feature of all 22 channels	did screening discard something useful?

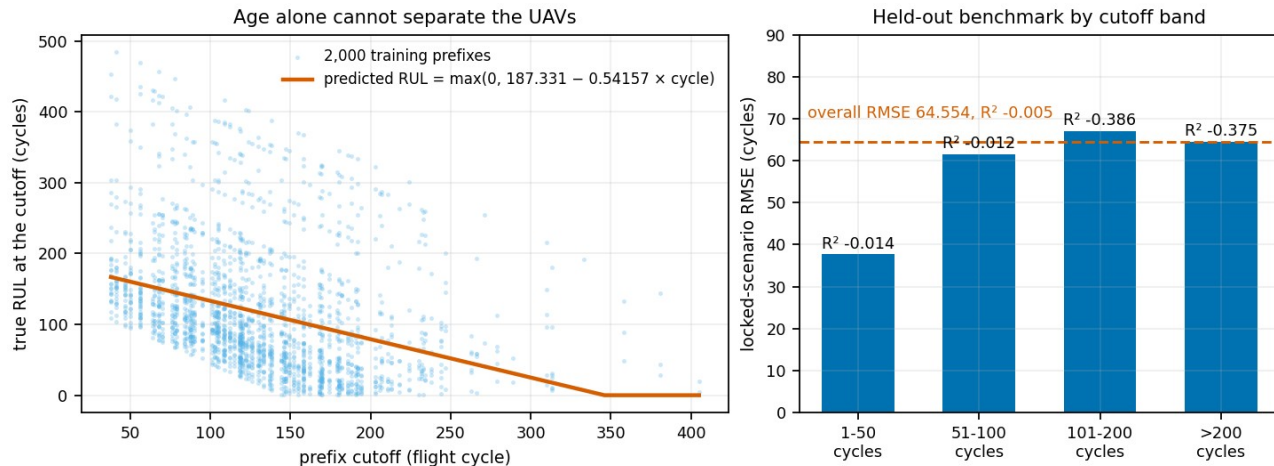
Fold-fitted scaling: value minus the training-fold median, divided by IQR / 1.349. 4,405 features scaled by IQR, 293 by a standard-deviation fallback, 12 by unit fallback.

The final set is not chosen here. It is chosen on the inner folds.

Sources: 1_dataset_construction steps 5-7 · feature_catalog.csv and preprocessing_config.json · claims C69-C77

The benchmark: age alone predicts nothing

A weighted linear fit on the cutoff cycle, evaluated on the same 20 locked scenarios



Locked-scenario result

- $R^2 -0.005$ · RMSE 64.554 · MAE 51.912 cycles
- Bias +17.712 cycles: age alone systematically overpredicts.
- 95% UAV-bootstrap interval for R^2 : $[-0.153, +0.129]$.

Why it is worth the compute

- It is the only number that makes a later RMSE interpretable.
- A complex model failing to beat it points at leakage control, weak features or an implementation fault, not at the data.

How the interval is built

- 1,000 resamples of whole UAVs, never of rows, because each UAV appears in 20 scenarios and those rows are not independent.
- Seed 20260814, as for the folds and cutoffs.

Figure: generated from 1_dataset_construction/9_cycle_only_baseline artifacts · claims C78-C86

Decision 1: bounded scenarios and a fitting cap at 125

PE_2 paired RMSE improvement, mean-fold, blue ExtraTrees and orange XGBoost. Public: +0.30377

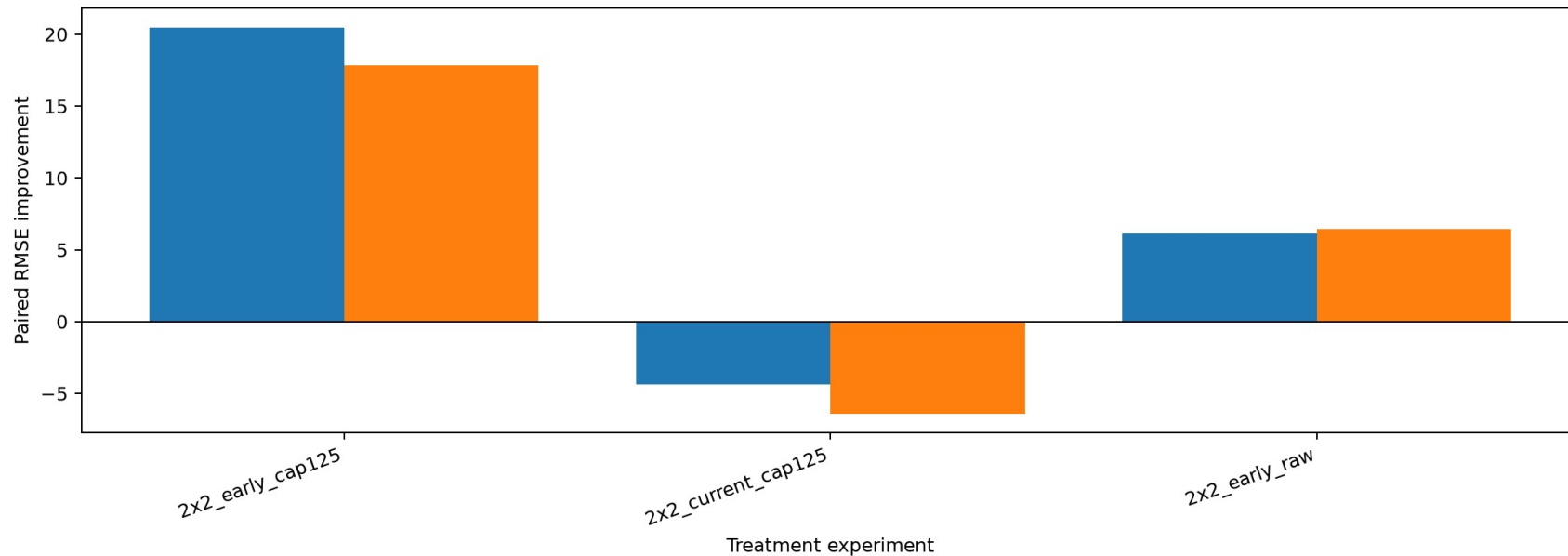


Figure: lower panel of the generated PE_2 figure · experiments/PE_2/runs/run_1 · claims C87-C94

Decision 2: pruned features and a calibrated tree blend

Locked architecture comparison, architecture study run 5, five held-out UAV folds. Public: +0.02012

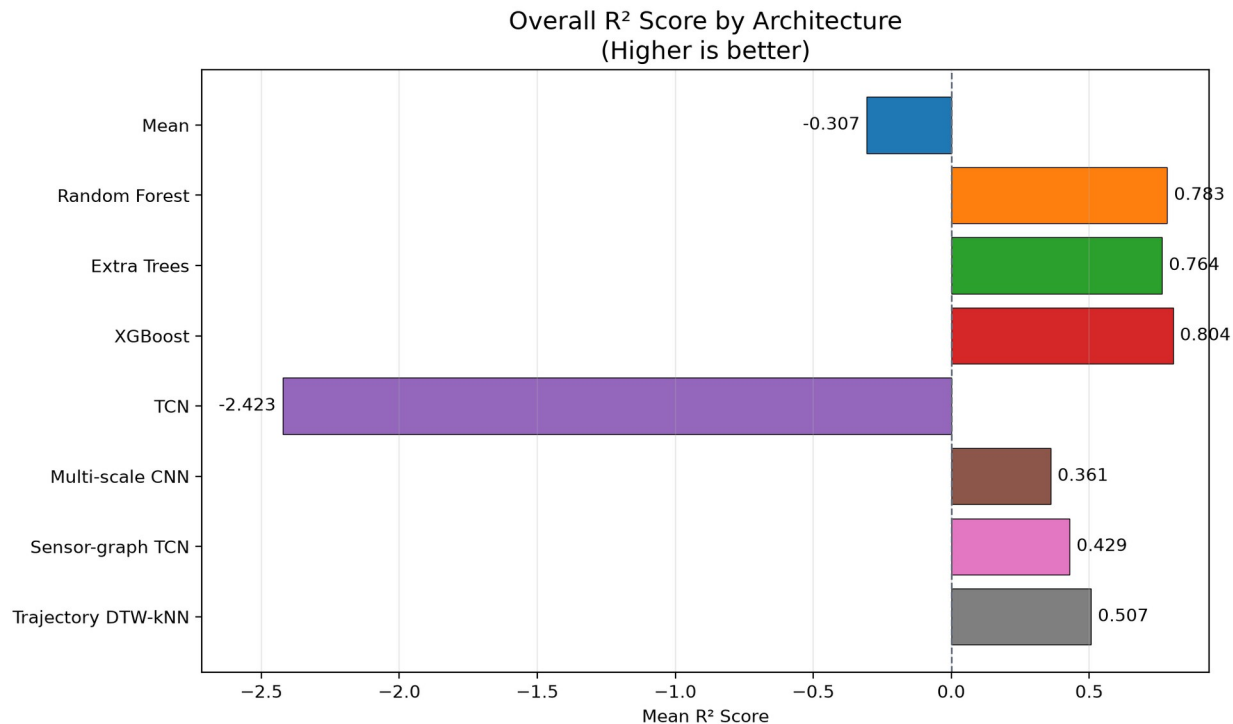


Figure: repository figure, unchanged · 2_model_architecture_study/runs/run_5 · claims C95-C103

Decision 3: conditional conservative calibration

PE_4: a quantile shift applied only where the model is likely to overpredict. Public: +0.00216

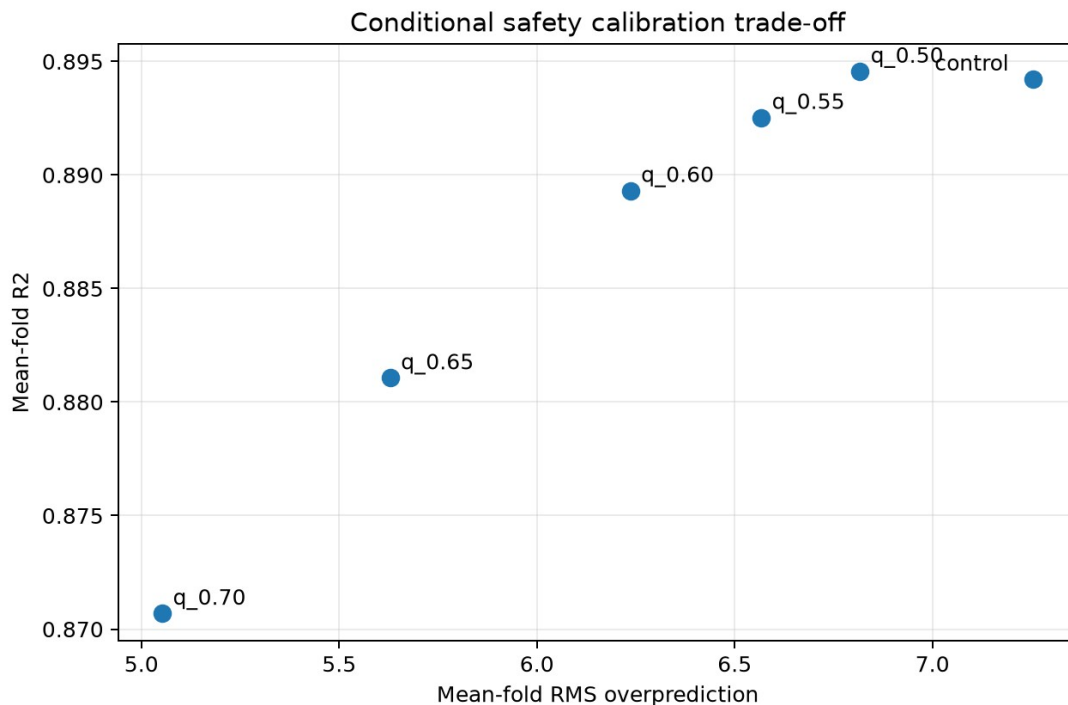
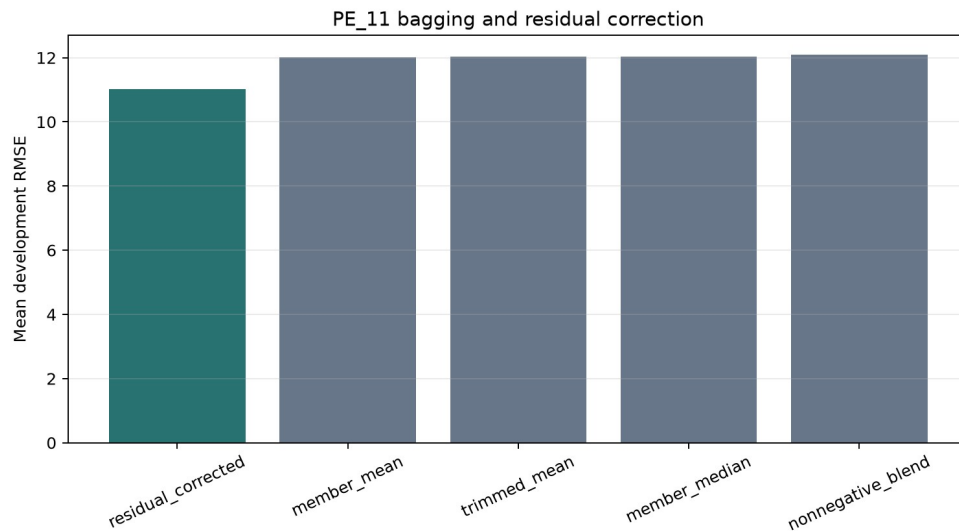
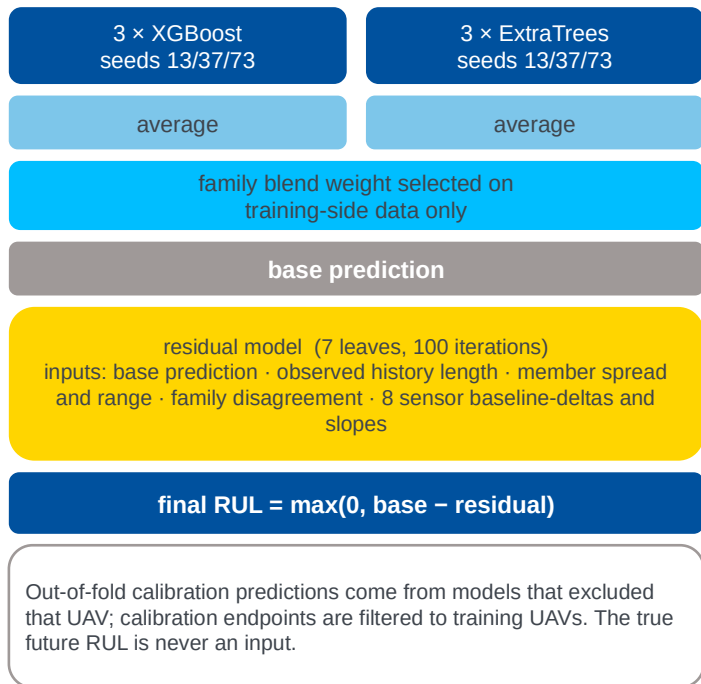


Figure: repository figure, unchanged · experiments/PE_4/runs/run_1 · claims C104-C108

Decision 4: a cross-fitted residual correction

PE_11: a small model that predicts the ensemble's own error from its disagreement. Public: +0.00911

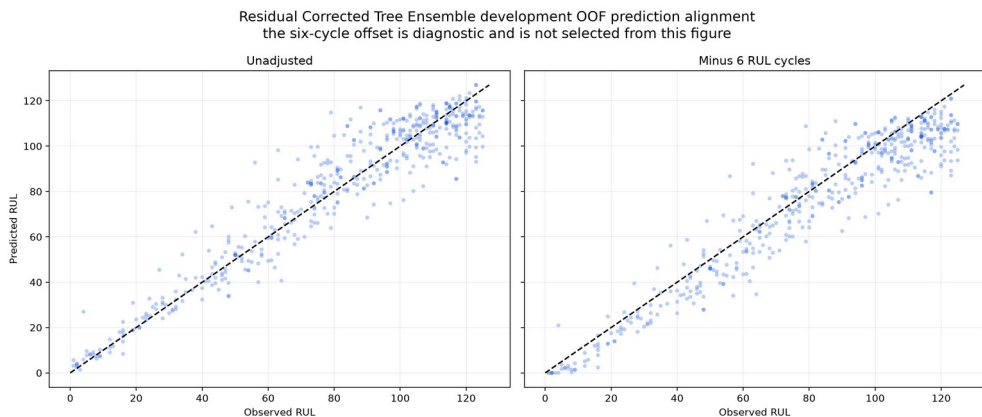


11.5080 → 11.0243 cycles, 4.20% better, 5 of 5 fold wins. Refitted by the Phase 3 Run 7 adapter on all 100 training UAVs.

Figure: repository figure, unchanged · experiments/PE_11/runs/run_1 · claims C109-C116

Why the chain stops at 0.87652

Run 7 development out-of-fold predictions, and every candidate evaluated since



Run 7 development out-of-fold predictions: the error is systematic, not random

After Run 7: six candidates, all gated

candidate	mean RMSE	wins	outcome
PE_20 nested refit	+1.56% worse	2/5	rejected
PE_18 TabPFN blend	-1.30%	4/5	to confirm
PE_21 two new seeds	-1.12%	6/10	rejected
PE_24 regime gate	-1.90%	11/15	rejected
PE_26 short history	-0.39%	6/15	rejected
PE_27 UAV subsets	screen only	3/5	stopped

Every promotion gate is declared before the run. Nothing passed, so nothing was submitted, so the public score is still 0.87652.

Figures and tables: [run_7/7_post_run_reporting](#) and [experiments/PE_15-PE_28](#) · claims C117-C127

How we reached 0.87652

Ordered by the submission that carried each decision

phase	decision	submissions	public R ² change
Phase 1 + 2	Bounded validation scenarios with a fitting cap at 125	Run 3 → 4	+0.30377
Phase 0 + 2	Drift-pruned 298 features, calibrated 50/50 tree blend	Run 4 → 5	+0.02012
Phase 2	Conditional conservative calibration at $q = 0.55$	Run 5 → 6	+0.00216
Phase 2	Cross-fitted residual correction on six seeded members	Run 6 → 7	+0.00911

Each row is a submission-to-submission difference, not an isolated ablation: a run can carry more than one change. Evaluation and target design dominated model choice.

What we do not claim

Development mean-fold R² is 0.90041; the recorded public score is 0.87652. Closing that needs about 10.0% lower RMSE on the same scored set, and no candidate after Run 7 passed a nested confirmation.

Sources: kaggle_scores.csv, pipeline_experiments.md, phase_0 and phase_1 artifacts · claims C128-C132

Not part of the 20-minute talk. Held for questions.

Backup slides

Backup: the complete Phase 0 broad review

Nine analyses, and the decision each one contributed

analysis	what it showed	what it decided
Cycle-wise fleet trends	6 channels flat across all cycles; late cycles have few surviving UAVs	removal shortlist; treat late-cycle trends cautiously
Descriptive statistics	channel medians span about 11 to about 50,000	robust scaling for scale-sensitive models; trees need none
Tukey extreme-value screen	~97% of extremes in 04/10/11 are isolated single cycles; 01 and 18 form long runs in 8-9 UAVs	spike flags vs regime flags; bounds fitted inside each fold
Row-level statistics	asymmetry concentrated in 07, 16, 18, 23, 25	robust scaling or Yeo-Johnson for linear and neural models
UAV-level statistics	01, 06, 12, 18, 26 differ strongly between UAVs	candidates for context and baseline features
Age-band statistics	14 channels change monotonically across the 1-50 / 51-100 / 101-200 / >200 bands	prioritised for temporal features; report metrics by band
Histograms and box plots	05, 07, 16, 18 are discrete or regime-like	operating-state features rather than smooth sensors
Flatline duration	20 and 27 are 100% flatlined; 03/08/14/17 above 92%; 07 at 96.5%	removal for the six; state features for 07 and 16

No channel was removed on distribution evidence alone. Skewness, extremes, flatlines and regimes are reasons to represent a channel differently; only UAV-grouped validation decides whether it earns its place.

Source: [literature_and_planning/development_documentation/phase_0_dataAnalysis.md](#)

Backup: redundancy, anomalies, and what was not removed

The strongest correlated pairs and the anomaly rates; every warning became an experiment

Strongest correlated pairs

pair	row r	UAV r	group
telemetry_19 / 21	0.963	0.999	1
telemetry_06 / 12	-0.839	-0.999	4
telemetry_13 / 28	-0.744	-0.995	3
telemetry_25 / 28	0.853	0.993	3
telemetry_16 / 22	-0.677	-0.992	3
telemetry_15 / 23	-0.954	-0.983	2

Groups: 1 = 19/21 · 2 = 15/23 · 3 = 13/16/22/25/28 · 4 = 06/07/11/12/24.

Fourteen of the twenty-two channels have a peer above 0.90 in at least one correlation view. None was removed here.

Anomaly diagnostics, training split

channel	extreme rows	UAVs	jump rows	UAVs
telemetry_26	8.45%	48	2.02%	8
telemetry_18	7.25%	9	19.48%	84
telemetry_01	5.28%	8	0.00%	0
telemetry_24	4.63%	99	3.54%	99
telemetry_10	1.80%	99	3.43%	99
telemetry_04	1.80%	99	3.41%	99

Nothing was deleted

Extremes in 04, 10 and 24 occur in almost every UAV, which reads as normal operating behaviour. Those in 01 and 18 sit in 8 and 9 UAVs, so they read as UAV-specific regimes. Persistent shifts appear only in 19 and 21, five UAVs each, and became candidate change-point features.

Source: 0_data_analysis/core_data_analysis/figures/feature_redundancy and figures/anomalies

Backup: the 27 features derived from every channel

All computed from cycles at or before the cutoff; verified by the prefix-causality assertion

group	features	what it tells the model
Age	$c, \log(1 + c)$	accumulated operating time
Level	x_1, x_c	start level and value at prediction time
Baseline	$\text{mean}(x_1..x_10), x_c - \text{baseline}$	movement from the UAV's own starting level
History	mean, SD, min, max, OLS slope	typical level, spread and long-term direction
Change	$x_c - x_(c-1), \text{mean and max } \Delta x $	abrupt change and typical volatility
Recent 5 / 20 / 50	mean, SD, slope, net change, deviation	short, medium and longer-term behaviour
State (07, 16)	unique values, transitions, transition rate, run length, time since change	operating mode, dwell time, flatlining

How causality is enforced

- Only cycles 1..cutoff enter any calculation.
- The raw files are never truncated; the cutoff is applied while computing.
- Feature names may not contain RUL, target, terminal, lifetime, final or future.

How it is verified

- 10 prefixes are recomputed after every post-cutoff telemetry value is replaced by an extreme artificial number.
- The prefix features must come out identical.
- Repeated for training, development, locked and test tables; all four are also checked for missing and non-finite values.

Source: 1_dataset_construction/5_prefix_feature_engineering and 6_feature_sets/artifacts/feature_catalog.csv

Backup: the cycle-only baseline, group by group

The same held-out predictions, reported by cutoff band, outer fold and terminal-lifetime quintile

By cutoff band

band	rows	R ²	RMSE	bias
1-50	80	-0.014	37.683	+4.980
51-100	340	-0.012	61.567	+11.320
101-200	1,020	-0.386	67.131	+38.872
>200	560	-0.375	64.577	-15.128

By outer fold (20 UAVs each)

fold	R ²	RMSE	MAE	bias
1	+0.040	55.570	46.943	+30.498
2	+0.032	57.025	49.669	+22.117
3	-0.068	76.307	59.860	+11.177
4	+0.020	70.106	53.041	+5.375
5	-0.055	61.306	50.046	+19.395

By terminal-lifetime quintile

quintile	R ²	RMSE	MAE	bias
1 shortest	-3.274	74.727	72.359	+72.359
2	-0.783	56.301	52.649	+52.649
3	+0.330	40.994	35.108	+33.089
4	+0.746	31.938	26.764	+1.286
5 longest	-0.059	96.859	72.680	-70.821

What the quintiles show

A cycle-only fit cannot be right for both short- and long-lived UAVs at once: it overpredicts the shortest by 72 cycles and underpredicts the longest by 71. Fold R² alone ranges from -0.068 to +0.040, which is why a single 80/20 split would not have been a measurement.

Source: [1_dataset_construction/9_cycle_only_baseline/artifacts/metrics/](#)

Backup: full locked architecture comparison

Architecture study run 5, five held-out UAV folds, identical folds and cutoffs for every entry

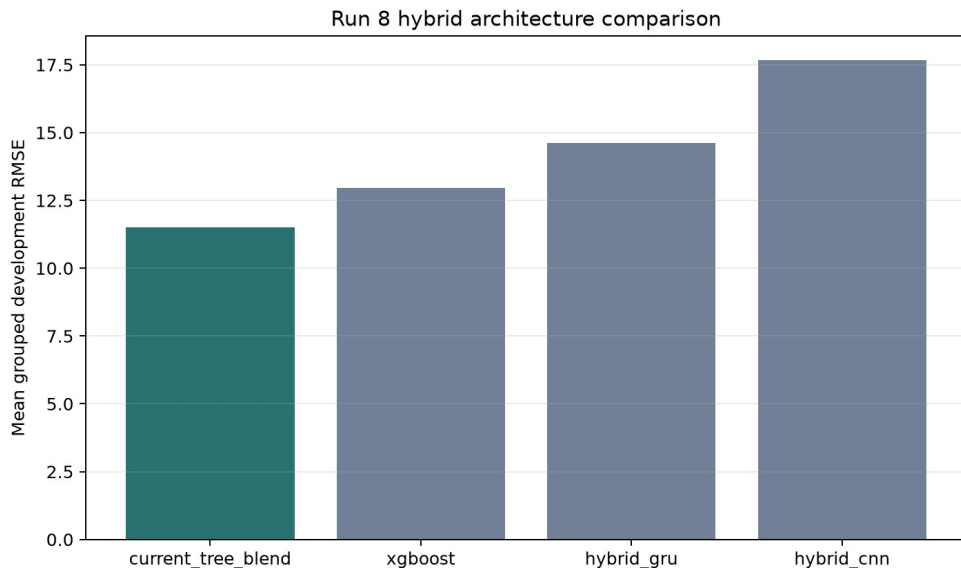
architecture	seeds	R ² mean	RMSE mean	R ² seed SD	R ² 95% UAV-bootstrap interval
XGBoost	3	0.8041	28.4989	0.0055	0.7528 - 0.8437
Random Forest	3	0.7827	30.0138	0.0030	0.7058 - 0.8348
Extra Trees	3	0.7640	31.2779	0.0027	0.6671 - 0.8372
Trajectory DTW-kNN	1	0.5067	45.2239	0	0.3804 - 0.6106
Sensor-graph TCN	3	0.4294	48.2278	0.1487	0.1602 - 0.6359
Multi-scale CNN	3	0.3607	51.4209	0.0622	0.2317 - 0.4691
Mean baseline	1	-0.3066	73.6023	0	-0.4848 - -0.1758
TCN	3	-2.4231	111.2088	2.1691	-9.7147 - 0.6382

Sequence models were given the same protocol, the same prefixes and a tuned budget.

Source: 2_model_architecture_study/runs/run_5/7_architecture_comparison/architecture_comparison.csv

Backup: sequence and hybrid models, in full

Architecture study run 8: hybrid combinations of a tree ensemble with a sequence encoder



Architecture study run 8, early/middle scenarios and cap-125 target. Neither hybrid won a single fold out of five.

Source: 2_model_architecture_study/runs/run_8

Dedicated temporal study (run 7)

model	mean R ²	mean RMSE
LSTM	0.6531	19.570
GRU	0.6424	19.832
TCN	0.5969	21.084
Multi-scale CNN	0.5325	22.712

Its own frozen sampling policy and lookback 20, so these values are not on a common scale with the left panel. The gate it had to clear was about R^2 0.89 and RMSE 10.7.

State the finding for the architectures, representations and budgets tested here — not as a claim about deep learning and RUL in general.

Backup: the complete cap and scenario matrix

PE_2, all four cells, with the target policy of each cell stated explicitly

scenarios	fitting target	model	inner RMSE	fold SD	inner R ²	bias	overpred. rate
current	raw	XGBoost	29.8059	1.7513	0.7708	+5.2165	67.35%
current	raw	ExtraTrees	32.7578	2.6545	0.7264	+4.4906	66.00%
current	cap 125	XGBoost	36.2141	2.5645	0.6690	-10.8488	46.70%
current	cap 125	ExtraTrees	37.1100	2.5884	0.6525	-10.7370	50.95%
early/middle	raw	XGBoost	23.3667	1.5326	0.4867	+11.4973	71.35%
early/middle	raw	ExtraTrees	26.6406	2.9202	0.2703	+10.6021	67.60%
early/middle	cap 125	XGBoost	11.9746	0.9418	0.8661	-0.9007	47.30%
early/middle	cap 125	ExtraTrees	12.2693	0.8427	0.8598	-1.0864	47.15%

Cells with different evaluation targets are not on a common R² scale. Fitting cap, evaluation support and prediction clipping are three different things.

Source: experiments/PE_2/runs/run_1/reporting

Backup: residual correction, calibration and the safety table

PE_11 and PE_4 in detail, with the nested-calibration guarantee stated in full

What the Run 7 adapter does

- Splits training UAVs internally into four groups.
- Fits the six tree members without the held group.
- Predicts it; repeats, so no UAV is seen by its own model.
- Filters calibration endpoints to training UAVs and balances weight per UAV.
- Fits the residual head there, then predicts the external held-out UAVs once.
- The earlier PE_11 report is not a fully nested estimate of its own stack; the Run 7 adapter does not share that dependency.
- Its remaining limitation: configurations and the workflow were chosen on development data now examined many times.

Conditional calibration policies (PE_4)

policy	mean-fold R^2	RMSE	RMS overpred.
control	0.89422	10.5306	7.255
q = 0.50	0.89456	10.5354	6.817
q = 0.55	0.89253	10.6430	6.567
q = 0.60	0.88931	10.8144	6.236
q = 0.65	0.88106	11.2152	5.629
q = 0.70	0.87070	11.6989	5.053

Rule declared in advance: only policies within 0.005 mean-fold R^2 of the best are eligible; among those, minimise RMS overprediction, then overprediction rate, then RMSE.

A later uncertainty-scaled variant improved near-failure RMS overprediction by 11.31% but raised RMSE by 1.71% and was rejected.

Source: experiments/PE_11 and PE_4, runs/run_1/reporting

Backup: representation experiments that were rejected

Every row is a matched development comparison whose control was retained

representation tested	matched development result	decision
Signal compression (median / PCA indices)	R^2 fell by roughly 0.09-0.11 in both families	keep individual features
Dense training prefixes (stride 5)	R^2 -0.0338 XGBoost, -0.0362 ExtraTrees; lost every fold	keep 20 test-like cutoffs
Per-UAV robust normalisation	combined set: +0.0051 XGBoost, -0.0065 ExtraTrees	keep raw features
Failure-cycle target instead of RUL	ΔR^2 -0.4002 XGBoost, -0.5073 ExtraTrees; 0/5 folds	keep RUL target
Personalised degradation-onset targets	XGBoost RMSE 12.15 → 19.59-19.69	keep cap 125
Drift-feature pruning	domain AUC 0.8565, but no treatment passed the accuracy gate	keep all features
Population degradation features	mean RMSE +2.5% (HGB) and +11.8% (ridge)	keep control

PE_28 now revisits the representation question deliberately: ten feature sets from 50 to 310 features crossed with two model recipes, varying sensor coverage, summary complexity, baseline averaging and window length one factor at a time. Screening uses grouped seed 20270107; confirmation uses seeds 20270117 and 20270127 on the same UAV population, so it is a robustness check rather than an untouched holdout.

Source: experiments/PE_5, PE_6, PE_7, PE_9, PE_13, PE_22, PE_28

Backup: uncertainty, endpoint duplication and score provenance

How the intervals are built, what the repeated endpoints do, and where each score comes from

What the bootstrap covers

- 1,000-3,000 resamples of whole UAVs, never of rows.
- Run 7 pooled R^2 interval 0.87779 - 0.92772.
- Run 7 minus Run 6 RMSE interval -1.252 to +0.446 cycles.
- It resamples fixed saved predictions, so it excludes training and model-selection uncertainty.
- A positive mean with an interval crossing zero is uncertain evidence, not proof of no effect.

Repeated endpoints

- 500 development rows contain 444 distinct (UAV, cutoff) pairs.
- Removing duplicates changes pooled R^2 only from 0.90446 to 0.90494.
- Duplication therefore does not explain the headline number, but it does affect weighting and precision.
- The UAV, not the endpoint, stays the resampling unit.

Public-score provenance

- Public scores were transcribed from user-supplied Kaggle screenshots recorded on 7 September 2026, not from a live lookup.
- The public leaderboard uses about 30% of the test data; the final standing uses the other 70%.
- The 0.87877 figure for the alternative script is user-reported and has no submission artifact in this repository.

Source: r2_research_2026_09_07/ and kaggle_scores.csv